

Nicholas Eisenberg, Ph.D.

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Experience

Senior Scientist - Machine Learning

June 2022 - Present

Nevada National Security Site

- **LLM Inference Server (Piccata):** Architected, deployed, and administered a multi-user LLM platform on a RHEL GPU server, exposing internally hosted models through OpenAI-compatible vLLM APIs. Built a FastAPI gateway for authentication, routing, usage tracking, and persistence. Developed `libserverctl`, a Python library for user, API-key, certificate, and usage administration, and built a `Streamlit` portal for access requests and provisioning API tokens and client certificates. Deployed the platform with `Podman`, `Nginx` reverse proxying and TLS/mTLS.
- **LLM-Assisted Image Annotation and Retrieval (Photo-Pilot):** Designed and deployed a `Streamlit` application that enables users to annotate and tag images with the assistance of multimodal LLMs hosted on Piccata. The annotations and metadata were persisted in a SQL database, and the portal allowed the users to query the resulting image collection. Deployed the application on a RHEL server using `Podman`, with `Nginx` providing reverse proxying.
- **Distributed Machine Learning / HPC:** Developed an internal PyTorch training framework with callback support and simplified configuration for Data Parallel (DP) and Distributed Data Parallel (DDP) training. Used the framework to refactor scientific ML codebases for distributed execution on Linux HPC clusters with SLURM, enabling researchers to scale model training across multiple GPUs.
- **AWS GPU Infrastructure Administration:** Built and administered an eight-GPU AWS EC2 instance for scientific ML workloads. Responsibilities included maintaining the GPU software stack, provisioning and managing EBS storage and filesystems, and administering Linux user accounts and access.
- **Internal Scientific Software Monorepo:** Maintained a large internal monorepo which supported ML and scientific applications, including shared Python libraries, dependency and release workflows, automated testing, CI/CD, and Linux deployment infrastructure.

Technical Skills

Programming Languages: Python, Bash, Lua, SQL, Cypher, $\LaTeX$

Libraries: PyTorch, Pandas, FastAPI, SQLAlchemy, Streamlit, Matplotlib, vLLM

Tools / Infrastructure: Linux (RHEL, Ubuntu), Podman, Docker, Nginx, Git, CI/CD, SLURM, AWS

Open-Source Projects

iron.nvim - Sole Maintainer

May 2024 - Present

github.com/Vigemus/iron.nvim

Sole maintainer of iron.nvim, a cross-platform Neovim plugin written in Lua for interactive REPL integration with 1,400+ GitHub stars. My responsibilities include feature development, code review, testing, issue triage, and technical support across Linux, macOS, and Windows.

Education

Auburn University - Ph.D.

June 2022

Research Area: Stochastic Partial Differential Equations

Advisor: Le Chen

University of Nevada, Las Vegas - B.S./M.S.

December 2015 / January 2018

Applied Mathematics

Publications

- 1: Eisenberg, N., Chen, L. *Interpolating the stochastic heat and wave equations with time-independent noise: solvability and exact asymptotics*. Stoch PDE: Anal Comp (2022).
- 2: Eisenberg, N., Chen, L. *Invariant Measures for the Nonlinear Stochastic Heat Equation with No Drift Term*. J Theor Probab (2024).